# PaN-Finder

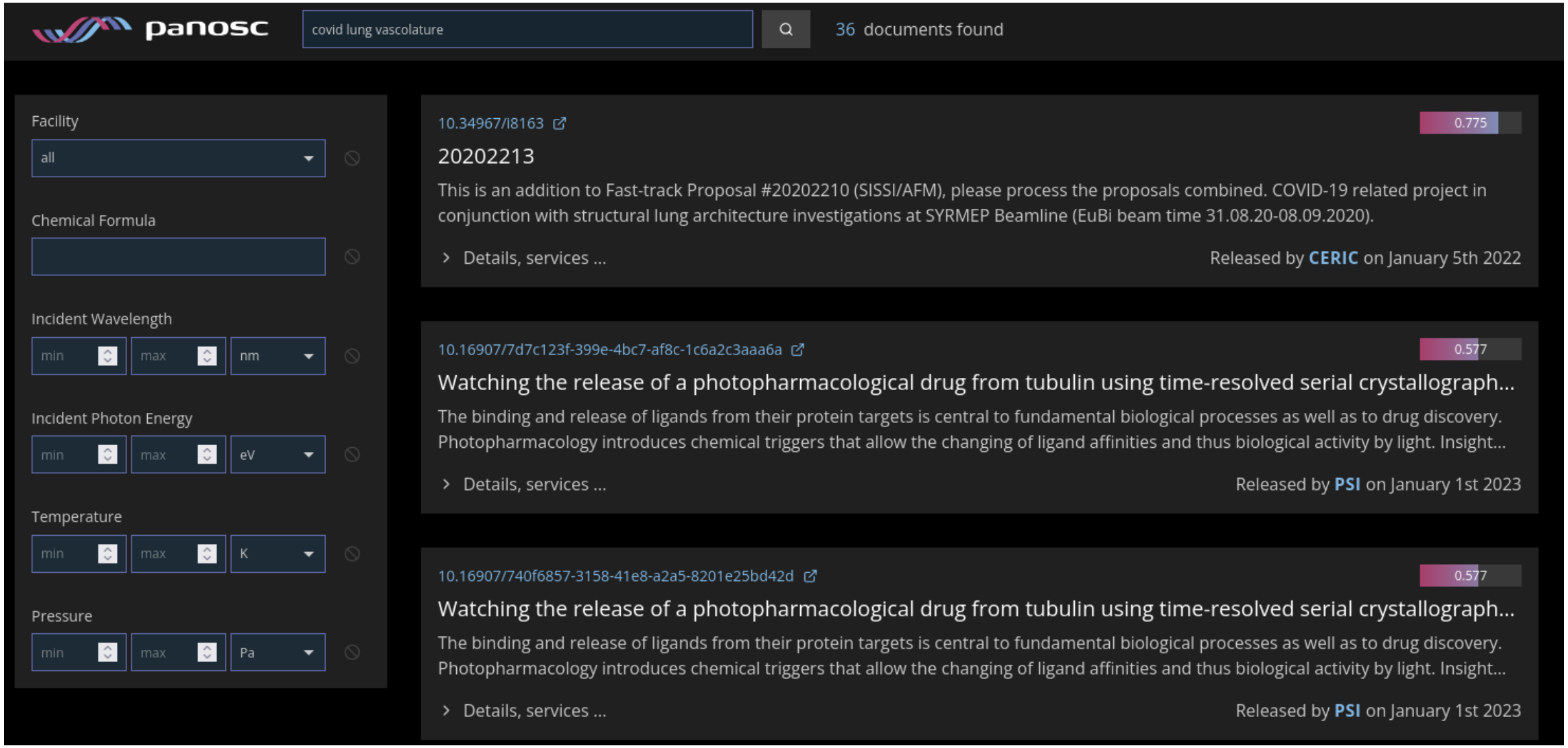
Max Novelli and the SIMS team (DMSC - ESS)

DAPHNE4NFDI lecture series, 2025-10-28

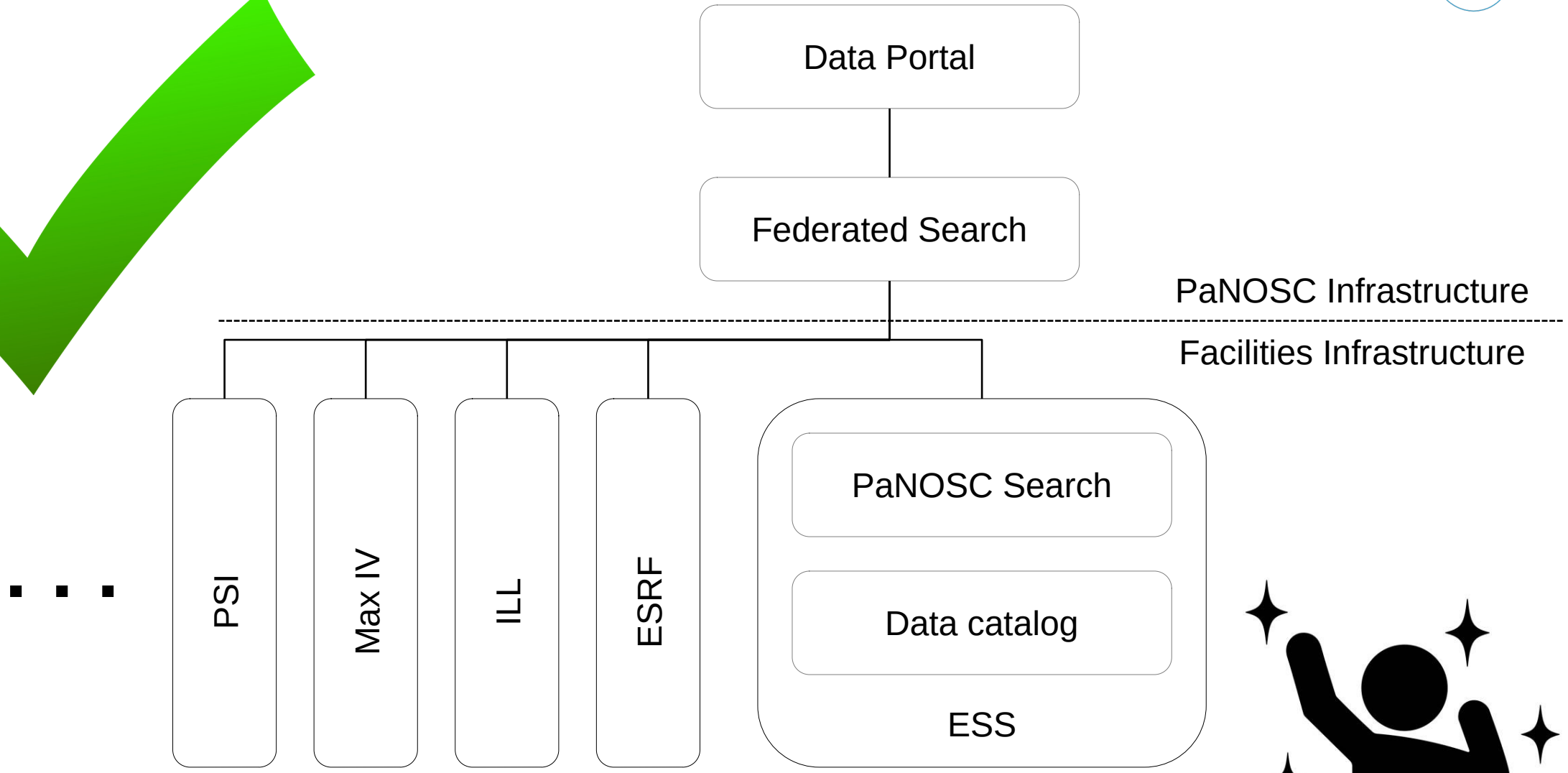
# The beginning

Greg Razoky on Unsplash

# PaNOSC Data Portal



# Success!!!


Data Portal
Federated Search
PaNOSC Infrastructure
Facilities Infrastructure
. . .
PSI
Max IV
ILL
ESRF
PaNOSC Search
Data catalog
ESS

# Few Challenges

- The project ended.
  Supported as in-kind contribution

- Connection between federated and local search APIs is not robust

- Local search APIs are hard to monitor

- Data curation at each facility varies a lot

# User Experience

panosc

covid lung vascolature

36 documents found

Facility

all

Chemical Formula

Incident Wavelength

min max nm

Incident Photon Energy

min max eV

Temperature

min max K

Pressure

min max Pa

10.34967/i8163 0.775

20202213

This is an addition to Fast-track Proposal #20202210 (SISSI/AFM), please process the proposals combined. COVID-19 related project in conjunction with structural lung architecture investigations at SYRMEP Beamline (EuBi beam time 31.08.20-08.09.2020).

Details, services ... Released by **CERIC** on January 5th 2022

10.16907/7d7c123f-399e-4bc7-af8c-1c6a2c3aaa6a 0.577

Watching the release of a photopharmacological drug from tubulin using time-resolved serial crystallograph...

The binding and release of ligands from their protein targets is central to fundamental biological processes as well as to drug discovery. Photopharmacology introduces chemical triggers that allow the changing of ligand affinities and thus biological activity by light. Insight...

Details, services ... Released by **PSI** on January 1st 2023

10.16907/740f6857-3158-41e8-a2a5-8201e25bd42d 0.577

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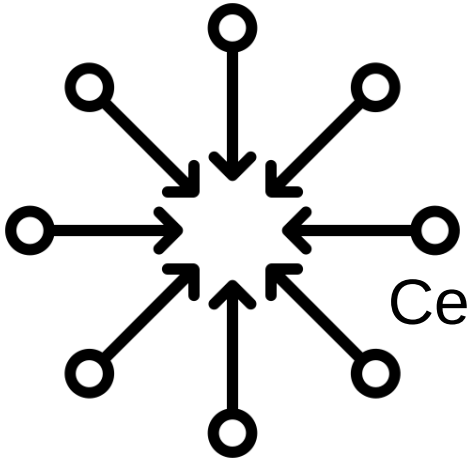
# The Vision

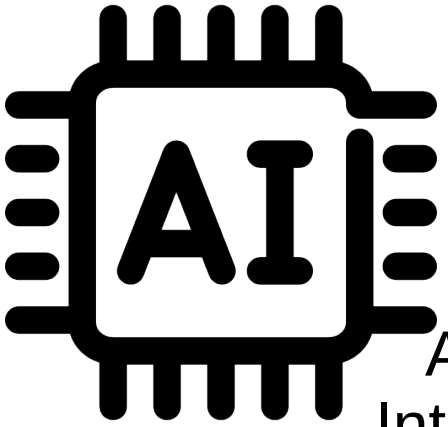
# The requirements



Centralized

Asynchronous data collection

Natural Language




Artificial Intelligence

Retrieval Augmented Generation

# The idea

I'm writing a new proposal for a new experiment to study the changes in molecular structure of a new synthetic carbon based material bonding with human blood red cells. I'm interested in running the experiments at a normal human body temperature. Can you show me related publications and similar proposals? Have similar experiments already been conducted?

In-vivo experiments with synthetic carbon based materials has been already performed and you can find information about them in the following resources:

- Resource 1 (URL 1)
- Resource 2 (URL 2)

I have not found information regarding new synthetic carbon based materials bonding with human blood red cells, but relevant information might be found in the following resources:

- Resource 3 (URL 3)
- Resource 4 (URL 4)

You also might want to check the following open datasets as they have been acquired from human blood red cells at normal body temperature:

- Dataset 1, RI, DOI
- Dataset 2, RI, DOI
- Dataset 3, RI, DOI

# The opportunity

OSCARS

Open Science Clusters' Action for Research & Society

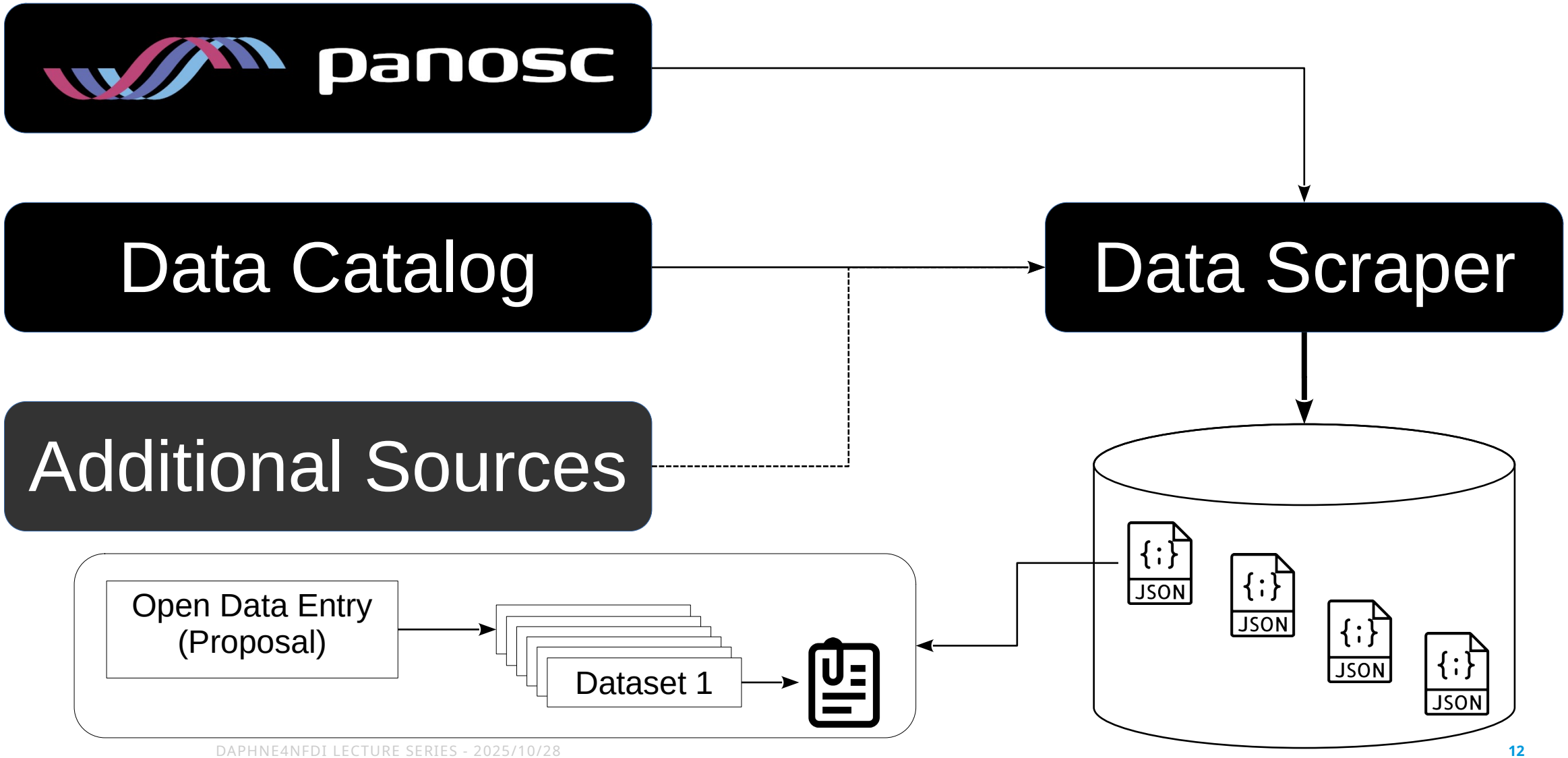
https://oscars-project.eu/

https://oscars-project.eu/projects/pan-finder-photon-and-neutron-federated-knowledge-finder

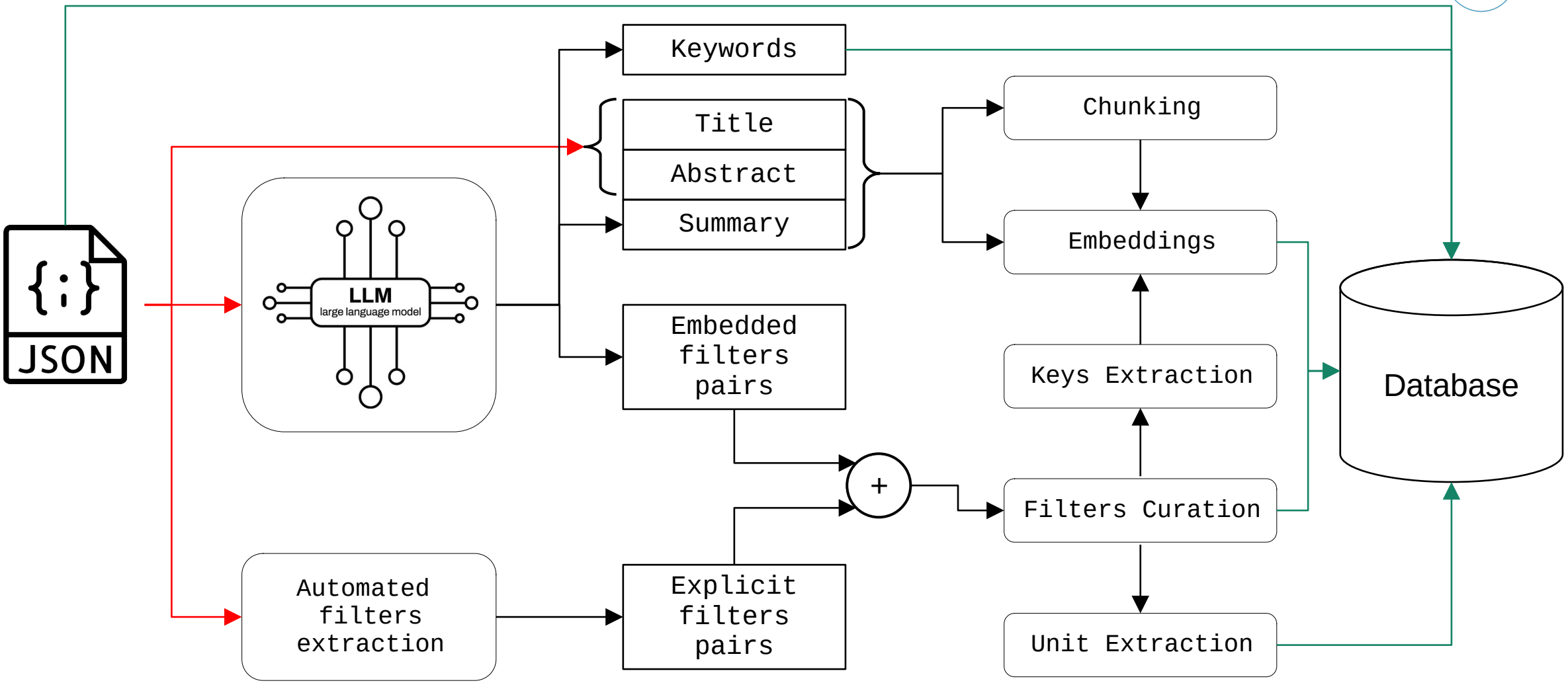
# Implementation

https://lifebalancestudio.co.uk/the-reality-of-yoga-poses/

# Data Collection

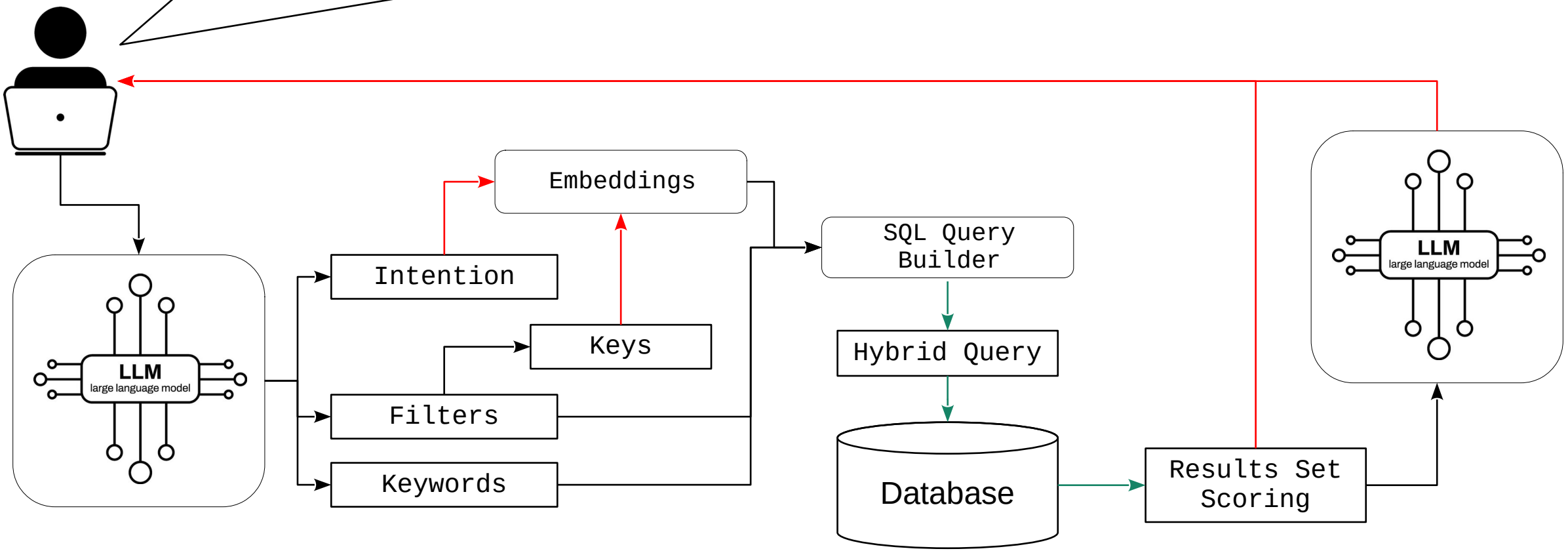


# Data Ingestion



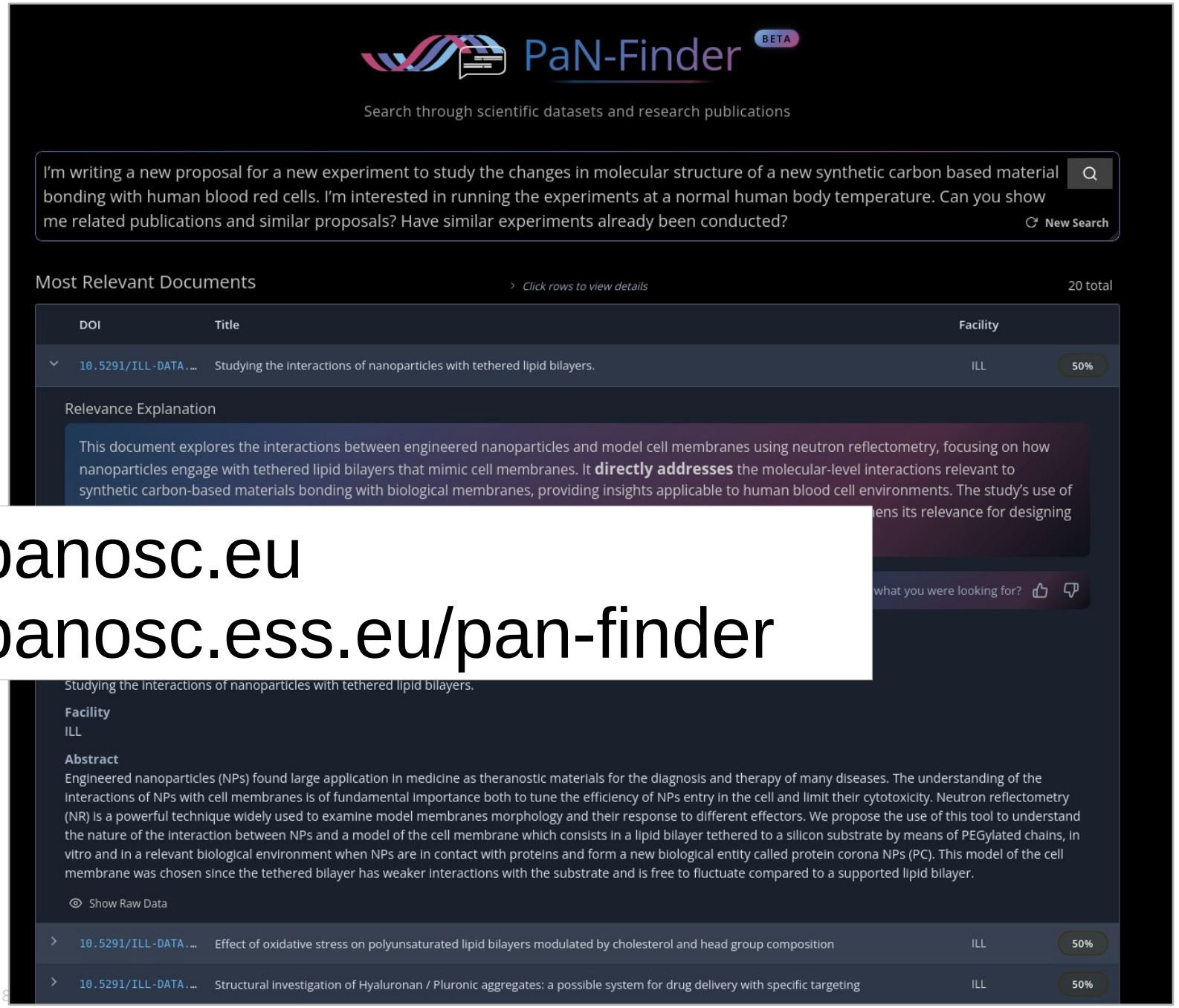
# User journey

I'm writing a new proposal for a new experiment to study the changes in molecular structure of a new synthetic carbon based material bonding with human blood red cells. I'm interested in running the experiments at a normal human body temperature. Can you show me related publications and similar proposals? Have similar experiments already been conducted?



# Beta version

https://pan-finder.panosc.eu
https://pan-finder.panosc.ess.eu/pan-finder



# User Prompt Components

I'm writing a new proposal for a new experiment to study the changes in molecular structure of a new synthetic carbon based material bonding with human blood red cells. I'm interested in running the experiments at a normal human body temperature. Can you show me related publications and similar proposals? Have similar experiments already been conducted?

New Search

```
{
  "intention": "publications and proposals on molecular structure of synthetic carbon based material bonding with human blood red cells",
  "keywords": [
    "molecular structure",
    "synthetic carbon based material",
    "human blood red cells"
  ],
  "filters": {
    "logic": "AND",
    "conditions": [
      {
        "name": "temperature",
        "operator": "=",
        "value": 37,
        "unit": "°C"
      }
    ]
  }
}
```

# Demo

# Query Examples

Query #1

Find datasets where lungs from a covid patient where scanned

DOI: 10.15151/ESRF-ES-436648953

Query #2:

Can you find me documents where elephants are equal to apples

DOI: ?????

Bonus Query:

1) Find reflectivity datasets on small samples, surface < 1cm2
2) Find reflectivity datasets on small samples with surface < 1cm2
3) Find reflectometry datasets on small samples with surface < 1cm2
4) Find datasets on small samples with surface < 1cm2 acquired with reflectometry technique

DOI: 10.15151/ESRF-ES-1824853946

# Query #1: prompt

find datasets where lungs from a covid patient where scanned

New Search

Relevance Explanation

**Most Directly Related Results**

- The study *Insight into effects of COVID-19 on human heart and lung tissue at sub-cellular level* investigates lung tissue from deceased COVID-19 patients using advanced imaging techniques, providing data that **fully aligns with** your request for scanned lung datasets from COVID-19 patients. (DOI: 10.15151/ESRF-ES-436648953)
- The paper *Multiscale Quantification of Covid-19's impact on lung vasculature from whole lobe to alveolar/microvascular scales* offers a comprehensive analysis of COVID-19's effects on lung structure at multiple scales, using imaging data from patient lungs, and **directly satisfies all aspects of** your query about scanned COVID-19 lung datasets. (DOI: 10.15151/ESRF-ES-471183885)
- Another entry titled *Multiscale Quantification of Covid-19's impact on lung vasculature from whole lobe to alveolar/microvascular scales* appears to be a duplicate or closely related study, further emphasizing the availability of detailed imaging data from COVID-19 patient lungs. (DOI: 10.15151/ESRF-ES-421386909)
- A third listing under the same title, *Multiscale Quantification of Covid-19's impact on lung vasculature from whole lobe to alveolar/microvascular scales*, reinforces the presence of datasets focused on scanned lung tissue from COVID-19 patients. (DOI: 10.15151/ESRF-ES-189558792)

# Query #1: Results

Most Relevant Documents

> Click rows to view details

20 found

| DOI | Title | Facility |
|---|---|---|
| 10.15151/ESRF-ES-436648953 | Insight into effects of COVID-19 on human heart and lung tissue at sub-cellular level | ESRF |
| 10.15151/ESRF-ES-471183885 | Multiscale Quantification of Covid-19's impact on lung vasculature from whole lobe to alveolar/microvascular scales | ESRF |
| 10.15151/ESRF-ES-421386909 | Multiscale Quantification of Covid-19's impact on lung vasculature from whole lobe to alveolar/microvascular scales | ESRF |
| 10.15151/ESRF-ES-189558792 | Multiscale Quantification of Covid-19's impact on lung vasculature from whole lobe to alveolar/microvascular scales | ESRF |
| 10.16907/ef0cfe43-0525-4e80-b77e-d6c9… | Airspace diameter map - a quantitative measurement of all pulmonary airspaces to characterize structural lung diseases | PSI |
| 10.16907/7eb141d3-11f1-47a6-9d0e-76f8… | Micrometer-resolution X-ray tomographic imaging of a complete intact post mortem juvenile rat lung | PSI |
| 10.5291/ILL-DATA.9-13-653 | Structural changes to the air-water interface following cholesterol oxidation in mixed lung lipid monolayers | ILL |
| 10.15151/ESRF-ES-845373199 | Structural characterization of cardiac, coronary and carotid macro-/micro-thrombi in COVID-19 positive and negative patients by SAXS and WAX | ESRF |
| 10.5291/ILL-DATA.9-13-843 | IN-SITU STUDY OF THE EFFECT OF GRAPHENE OXIDE NANOPARTICLES ON THE SURFACE PROPERTIES OF DIPALMITOYLPHOSPHATIDYLCHOLINE | ILL |
| 10.5291/ILL-DATA.8-04-859 | Understanding sickle cell disease and its genetic cure from the structure of haemoglobin | ILL |
| 10.15151/ESRF-ES-1351926332 | Effects of novel therapies for pulmonary arterial hypertension - do they exert the effect we think they do? | ESRF |
| 10.15151/ESRF-ES-562896665 | Fast scanning SAXS of hydrated biological cells | ESRF |
| 10.5291/ILL-DATA.8-03-775 | Investigating the solution conformation and mechanism of action of antimicrobial alginate oligosaccharides (OligoG). | ILL |

# Query #1: details

10.15151/ESRF-ES-436648953 | Insight into effects of COVID-19 on human heart and lung tissue at sub-cellular level | ESRF

**DOI**

10.15151/ESRF-ES-436648953

**Title**

Insight into effects of COVID-19 on human heart and lung tissue at sub-cellular level

**Facility**

ESRF

**Summary**

The study aims to investigate the sub-cellular effects of COVID-19 on heart and lung tissues by analyzing autopsy samples from deceased patients. Using X-ray cryo-nano-holotomography, researchers seek to uncover the 3D structural changes in these vital organs, which remain poorly understood. The goal is to enhance knowledge of the tissue damage caused by terminal COVID-19 infection.

**Content**

COVID-19 affects multiple vital organs and leads to drastic changes in tissue architecture not only in the lung but also in the heart. The understanding of these alterations is currently very limited and no high resolution 3D data is available. We propose to investigate damage in heart and lung tissue from patients having died with COVID-19 and who have undergone autopsy. This will be achieved by X-ray cryo-nano-holotomography. The aim of the proposal is to reveal the 3D changes occurring in tis...

# Query #2: prompt and results

can you find me documents where elephants are equal to apples

New Search

Most Relevant Documents

0 found

**No research publications found**

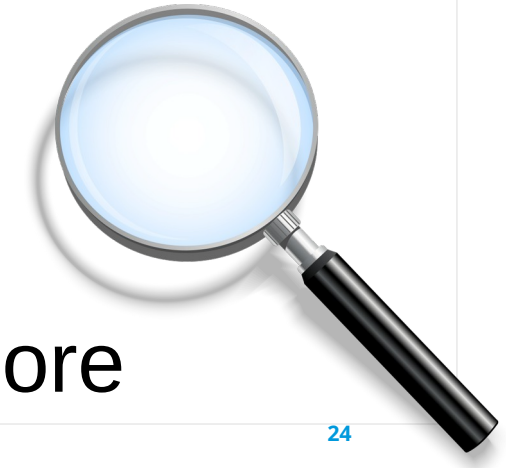
Your search didn't return any matching documents in our scientific database

# What’s next?

# Current and future work

- Benchmarking
- More Testing:
  - Different models (size, cost, performance)
  - Prompt Engineering
  - Model fine tuning
- Refine Natural Language Response
- Automating Data Collection
- Driving better data curation
- User feedback:

  1-on-1 sessions, answers validation, and more



# Thank you for your attention

Thanks to Janos Babik and the SIMS team

Questions?

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max.novelli@ess.eu